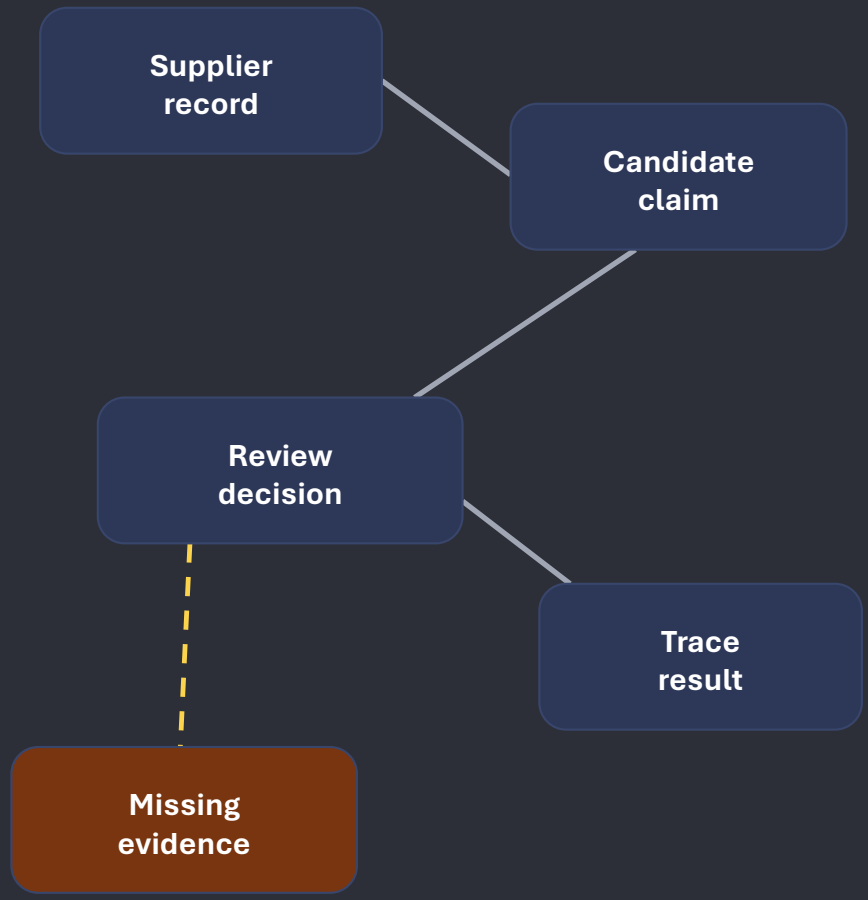
KLARROW

The evidence layer for food-traceability investigations

Turn fragmented records into a trace people can inspect, challenge and explain.

WORKING PROTOTYPE

Local Llama inference · Source-linked claims · Human-reviewed relationships



A trace is only as trustworthy as the evidence behind each link.

The investigation starts before the data is clean

A quality manager needs to reconstruct a path across records that were never designed to fit together.

Records are scattered

- **PDF invoices**
- Batch sheets
- Scanned dispatch notes

Identifiers drift

- **RM-0873**
- RM 0873
- RM-O873 (letter O)

The trail has gaps

- **320 kg received**
- 180 kg consumed
- Dispatch proof missing / unclear

When a supplier flags a lot, staff must identify affected batches and destinations without hiding contradictions or missing records.

No market-size, error-rate or time-saved claim is made here; these remain to be validated with practitioners.

An evidence-reconciliation workbench

Klarrow helps turn existing records into reviewable claims and a trace with visible uncertainty.

01

Ingest

Keep originals. Parse supported text and tables.

02

Extract

Local Llama proposes candidate facts with source evidence.

03

Review

Resolve identities and relationships with a human decision.

04

Trace

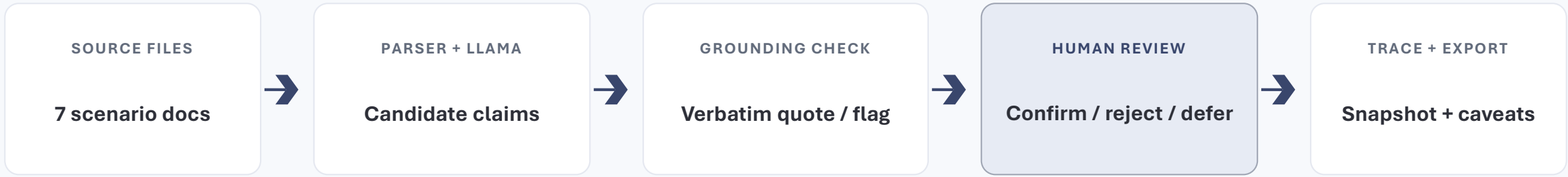
Separate confirmed scope, possible exposure, conflicts and gaps.

Positioning **Designed to work beside existing systems-not replace an ERP.**

That interoperability advantage is a product hypothesis; no ERP integration has been demonstrated.

AI proposes. Evidence and people control the trace.

A short, inspectable chain from a document to a relationship decision.



Guardrails in the current implementation

- AI facts remain unreviewed candidates.
- Unverifiable values can be flagged without a locator.
- A rejected or edited support fact can demote dependent links.
- Trace snapshots are stored and reloaded read-only.

Known capability boundary

- Text, CSV and spreadsheet extraction are supported.
- Scanned PDFs and images are retained, not OCR-read.
- A verbatim quote is not proof of semantic correctness.
- A human must verify claims before operational reliance.

FICTIONAL TEST SCENARIO

One authored scenario. A complete investigation path.

Karoo Crunch Foods receives an alert linked to raw-material lot RM-0873.

Input evidence

- Supplier recall alert
- Goods-received note
- Stock ledger + batch sheets
- Two dispatch notes

7 authored documents · SHA-256 recorded

Extraction results

13 AI candidate facts

8 quote-verbatim at cited line

5 identifier-only; flagged

35 parser facts · 48 total

Human gate

8 relationship proposals

7 confirmed by reviewer

1 false-positive rejected

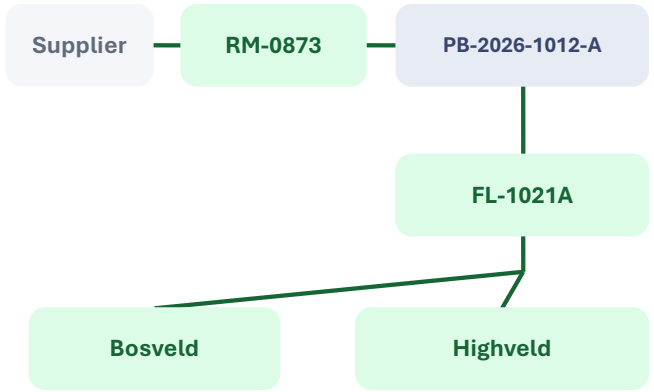
Alias RM-O873 was deferred, not silently merged.

Single authored scenario; extraction grounding is not a semantic accuracy score.

A trace that shows both what is supported and what is not

Final run after review: the false-positive customer relationship did not enter the confirmed trace.

Confirmed scope



1 finished lot · 2 customers · 5 confirmed trace edges

Unresolved evidence

INSUFFICIENT DATA

320 kg received

180 kg recorded consumed

6,600 units downstream

No justified kg-to-unit conversion; mismatch stays open.

Decision trace

- **FL-1034C**
  excluded after false-positive rejection
- **RM-O873**
  ambiguous alias deferred
- **3 findings**
  quantity mismatch, ambiguous ID, unreviewed claims

Possible scope was empty in the final snapshot; ambiguity remains disclosed in findings.

A reviewable evidence layer-not an autonomous recall decision

The intended advantage is preserving ambiguity and provenance across the investigation.

What is demonstrated in code

- Local Llama can produce candidate facts from supported text records.
- Quotes are checked against cited source lines; unverifiable identifier matches are flagged.
- Relationships require explicit human decisions.
- Rejected/edited supporting claims can demote dependent links.
- Trace traversal and snapshots are deterministic.

What is still a hypothesis or limitation

- Faster than manual reconstruction - not measured.
- Better than existing traceability software - not validated.
- Robust on real business documents - not yet tested.
- ERP integrations - not implemented.
- Quote-verbatim does not prove the extracted value is semantically correct.

Human-owned food-safety decisions · explicit uncertainty · auditable review

Observed results-without pretending this is an accuracy study

Run 3 · isolated pilot data root · local llama3.2:3b · one authored fictional scenario

53.1 s	8 / 13	5 / 13	8 proposals
7-document extraction	verbatim quote at cited line	identifier-only; flagged	7 confirmed · 1 rejected

Outcome observed

- Confirmed trace: FL-1021A → Bosveld + Highveld.
- False-positive destination for FL-1034C rejected.
- RM-O873 alias deferred; not silently merged.
- Mass balance remains insufficient.

Do not overclaim

- All 48 facts remained unreviewed in this run.
- 4 quote-verified claims had values on adjacent lines.
- No manual baseline, customer study or accuracy rate.
- The 7 input documents were authored fiction.

Run timings are single-run observations, not a performance guarantee. The benchmark does not measure customer value.

Help us validate the investigation workflow in the real world

Klarrow is seeking expert review and supervised validation-not claiming traction it has not earned.

01

Practitioner review

A food-safety / QA professional to challenge the workflow and its report.

02

Authorised test records

Redacted or purpose-built records to test real document variability safely.

03

Measured comparison

One timed mock recall: manual process vs Klarrow, independently checked.

Klarrow — **The evidence-reconciliation workbench for food-traceability investigations.**

Next milestone: practitioner-reviewed benchmark + controlled pilot plan. Not yet a production or compliance tool.